Introduction to Mobile Development

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MODE 4201

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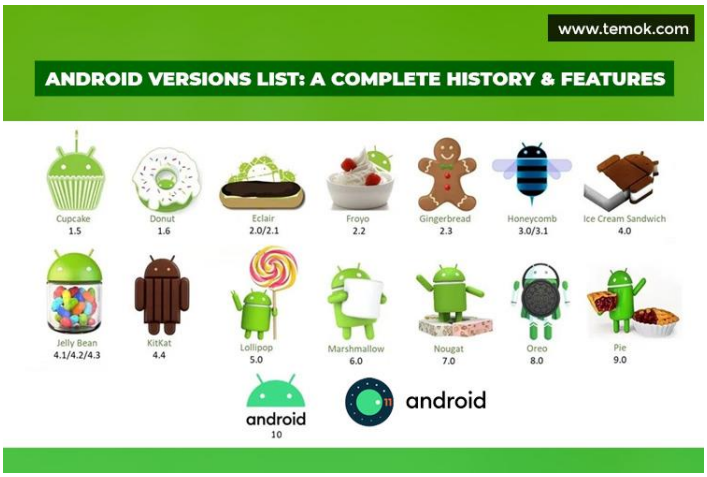
Mobile Development

Mobile Operating Systems

A mobile operating system is a software that allows users to use applications and programs on phones and tablets. The two most popular mobile operating systems are iOS created and developed by Apple which runs on their iPhones, and Android is created and developed by Google but is also open source allowing different companies to design an operating system for their specific models.



iOS was developed and released in 2007 on the first iPhone and has gone through many iterations throughout the years, spanning 16 major versions, with the most recent one announced this year. While android first released in 2008 and has gone through way more iterations due to different phones needing different versions, however the main version started at 1 and is now up 13. Which released in august this year.

Each operating system has advantages, iOS's advantages include the incredibly popular iMessage which is an iOS specific messaging app that allows Apple users to speak with each other over data. Allowing for fast messaging and great picture/video quality. Another advantage is every device that has iOS is

compatible with every Apple product. These advantages are not without their problems, iMessage does not communicate well with android phones for messaging causing poor picture/video quality between the two. Another disadvantage with iOS is the lack of customization within the iOS, you get what apple gives you.

Androids' advantages include fully customizing your UI on your device, make your phone do what you need and look how you want. Another advantage is Android runs on many different devices allowing more competition allowing you to get a phone with this operating system for a more affordable price. But this can lead to some disadvantages like the operating system runs slower on low specification devices, and the operating system is very large. Android is also more susceptible to virus as they do not offer better anti virus.

Mobile Device

Manufacturer: Samsung

Phone Model: Samsung Galaxy S22 Ultra 5G

Price: $1,639.99

Release Date: February 25th, 2022

Operating System: Android 12 with ONE UI 4.1

Chipset: Snapdragon 8 GEN 1

Processor: Octa-core (2x2.4 GHz Kryo 670 Gold & 6x2.0 GHz Kryo 670 Silver)

RAM: 12gb

Internal Memory: 256gb

External Memory: N/A

Display: 6.8" edge Quad HD+ Dynamic, AMOLED 2X,
Infinity-O Display (3088x1440),
500 ppi,
HDR10+ certified,
120Hz refresh rate

Rear Camera: Quad camera

1) 12MP Ultra Wide Camera
   Dual Pixel AF
   Pixel size: 1.4μm
   FOV: 120˚
   F.No (aperture): F2.2
   1/2.55" image sensor size
2) 108MP Wide-angle Camera
   PDAF, OIS
   Pixel size: 0.8μm (12MP 2.4μm)
   FOV: 85˚
   F.No (aperture): F1.8
   1/1.33" image sensor size
3) 10MP Telephoto Camera
   Dual Pixel AF
   Pixel size: 1.12μm
   FOV: 11˚
   F.No (aperture): F4.9
   1/3.52" image sensor size
   10x Optical Zoom

4)10MP Telephoto Camera
Dual Pixel AF
Pixel size: 1.12μm
FOV: 36˚
F.No (aperture): F2.4
1/3.52" image sensor size
3x Optical Zoom
Front Camera: 40MP Selfie Camera
PDAF
Pixel size: 0.7μm (10MP 1.4μm)
FOV: 80˚
F.No (aperture): F2.2
1/2.82" image sensor size

GPS: Yes

Battery: 5000 mAh

Battery Talk Time: 18 hours (No internet just used for calls)

Battery standby: 13 hours (Casual use)

Network: 5G

Synchronization with Desktop: Yes

Voice Enabled: Yes

Browser: Samsung Internet

Additional Feature 1: S-pen

Additional Feature 2: 100x Zoom

Mobile Plan

Service Provider: Telus

Plan: Unlimited 20

Calls: Unlimited Nationwide

Data: High speeds up to 20gb

Messaging: Unlimited nationwide texts, pictures, video messaging

Global Texting: Canada only

Voicemail: Advanced voicemail

Call features: Conference Calling, Call Display, Outbound Call Display, Call Waiting

Service Credits: Caller ID Outbound

Fees: $80

Usage Policy: N/A

Coverage: Nationwide

Additional Feature 1) No Data Overages

Additional Feature 2) Hotspot 20gb Data

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